$$\int_0^\infty K(s,t)\, f(t)\, dt$$

$$K(s,t) = e^{-st}$$

$$\mathcal{L}\{f(t)\} = F(s)$$

$$= \int_0^\infty e^{-st} \cdot f(t)\, dt$$

$$= F(s)$$

$$\mathcal{L}\{1\} = \boxed{\int_0^\infty e^{-st} \cdot 1\, dt}$$

$$= \frac{1}{s}$$

$$\mathcal{L}\{t\} = \int_0^\infty e^{-st} \cdot t \; dt$$

$$u = t$$
$$v = e^{-st}$$

$$\int uv \, dt$$

$$= u \int v \, dt - \int \left(\left(\frac{du}{dt}\right) \cdot \int v \, dt \right) dt$$

$$= t \cdot \frac{e^{-st}}{-s} \Bigg|_0^\infty - \int_0^\infty 1 \cdot \frac{e^{-st}}{-s} \, dt$$

$$= 0 + \frac{1}{s} \boxed{\int_0^\infty e^{-st} \; dt}$$

$$= 0 + \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$$

$$\boxed{m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = f(t)}$$

$$\frac{d^2x}{dt^2} = \text{Laplace domain?}$$

$$\Rightarrow f''(t)$$

$$\frac{dx}{dt} = f'(t)$$

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} \cdot f'(t)\,dt$$

$$= e^{-st} f(t)\Big|_0^\infty + s\boxed{\int_0^\infty e^{-st} f(t)\,dt}$$

$$= -f(0) + s \cdot F(s)$$

$f(0) = 0$

Assuming zero initial condi-tion

$$\mathcal{L}\{f'(t)\} = s \cdot F(s)$$

$$\mathcal{L}\left\{\frac{dx}{dt}\right\} = s \cdot F(s)$$

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - s f(0) - f'(0)$$

Assuming zero initial condition

$$\mathcal{L}\{f''(t)\} = s^2 F(s)$$

$$\mathcal{L}\{f'''(t)\} = s^3 F(s)$$

Take Laplace transform of the spring mass damper system

$$m \cdot s^2 \cancel{X}(s) + c \cdot s X(s) + k X(s) = F(s)$$

$$\Rightarrow X(s) \{ ms^2 + cs + k \} = F(s)$$

$$\Rightarrow \frac{X(s)}{F(s)} = \boxed{\frac{1}{ms^2 + cs + k}}$$

Transfer function

$$T(s) = \frac{1}{ms^2 + cs + k}$$

$$= \frac{1}{(s-a)(s-b)}$$

$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$$

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+2^2}$$

$$\mathcal{L}\{\cos 2t\} = \frac{s}{s^2+k^2}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+k^2}\right\} = \cos kt$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$\mathcal{L}^{-1}\left\{\frac{10}{s}\right\} = 10$$